

BWT2

Inorganic Chemistry

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A materials lab that recently earned ISO certification compares sodium, magnesium, and chlorine, all in period 3. Which has the largest atomic radius?

- A. Sodium
- B. Chlorine
- C. Magnesium
- D. All the same, since they fill the same shell.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which of these has the highest first ionization energy: Li, Na, K, or Cs?

- A. Cs
- B. Li
- C. All equal, since members of a group always share the same ionization energy.
- D. K

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A salt company that ships 300 tons a week sells sodium chloride. Which pair of properties fits this ionic compound?

- A. High melting point; conducts well as a solid.
- B. Low melting point; never conducts.
- C. High melting point; conducts when melted or dissolved, not as a solid.
- D. Dissolves in every solvent.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A semiconductor fab receives ammonia gas in 50-kilogram cylinders. The nitrogen in NH₃ has three bonds and one lone pair. What is the molecular shape?

- A. Tetrahedral
- B. Trigonal pyramidal
- C. Bent
- D. Trigonal planar

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A central atom has four bonding pairs and two lone pairs (AX₄E₂), as in XeF₄. What is the molecular shape?

- A. Octahedral
- B. Square planar
- C. Seesaw
- D. Tetrahedral

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A specialty-gas firm founded by two engineers runs the reaction $\text{BF}_3 + \text{NH}_3 \rightarrow \text{F}_3\text{B}-\text{NH}_3$. Which species acts as the Lewis acid?

- A. Both, since they share the new bond.
- B. Neither, since no H⁺ is transferred.
- C. BF₃
- D. NH₃

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A pigment maker whose products are sold in 40 countries uses K₄[Fe(CN)₆]. Cyanide is CN⁻. What are iron's oxidation state and coordination number?

- A. +3; 6
- B. +2; 4
- C. +6; 6
- D. +2; 6

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

In [Co(NH₃)₅Cl]Cl₂, cobalt is +3. What is the charge on the complex ion in brackets?

- A. +3
- B. 0
- C. +7
- D. +2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A mine's water-treatment plant, which handles 2 million gallons a day, removes iron with $\text{MnO}_4^- + 5\text{Fe}^{2+} + 8\text{H}^+ \rightarrow \text{Mn}^{2+} + 5\text{Fe}^{3+} + 4\text{H}_2\text{O}$. Which species is the oxidizing agent?

- A. MnO_4^-
- B. Fe^{2+}
- C. Mn^{2+}
- D. H^+

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

In the same reaction, manganese goes from +7 to +2. How many electrons does each manganese gain?

- A. 5
- B. 9
- C. 2
- D. 7

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A research group on a three-year grant studies $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, a high-spin octahedral d5 complex. How many unpaired electrons does it have, and what is its magnetism?

- A. 3; paramagnetic
- B. 0; diamagnetic
- C. 5; paramagnetic
- D. 1; paramagnetic

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A teacher whose school just remodeled its chemistry lab drops small pieces of lithium, sodium, and potassium into water behind a safety shield. Which reacts most vigorously, and why?

- A. Lithium, because smaller atoms react faster.
- B. All the same, since group members always react identically.
- C. Sodium, because it is in the middle.
- D. Potassium, because its outer electron is farthest from the nucleus and easiest to remove.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

~~A materials lab that recently earned ISO certification~~ compares sodium, magnesium, and chlorine, all in period 3. Which has the largest atomic radius?

Target word: largest atom in a period **Decides:** Radius decreases left to right as nuclear charge rises.

A	Correct	Leftmost in the period, lowest effective nuclear charge.
B	Inverted operation	Assumes more electrons across a period make the atom bigger.
C	Confusion set	Middle of the three; not the extreme.
D	Dropped element	Drops the rising nuclear charge that pulls the shell in.

2. Answer B Concept 1 • transfer

Which of these has the highest first ionization energy: Li, Na, K, or Cs?

Target word: reverse trend: ionization energy down a group **Decides:** Ionization energy decreases down a group.

A	Inverted operation	Largest radius, lowest ionization energy.
B	Correct	Smallest atom; its outer electron is held most tightly.
C	Absolute language	Group members share valence structure, not identical energies.
D	Confusion set	Lower in the group than Li.

3. Answer C Concept 2

~~A salt company that ships 300 tons a week~~ sells sodium chloride. Which pair of properties fits this ionic compound?

Target word: ionic properties **Decides:** Ions are locked in a lattice until melted or dissolved.

A	Adjacent category	Solid conduction is a metallic property.
B	Confusion set	Describes a molecular covalent substance.
C	Correct	Strong lattice forces; ions must be mobile to conduct.
D	Absolute language	Ionic solids dissolve in polar solvents, not in nonpolar ones.

4. Answer B Concept 3

~~A semiconductor fab receives ammonia gas in 50 kilogram cylinders.~~ The nitrogen in NH_3 has three bonds and one lone pair. What is the molecular shape?

Target word: molecular shape **Decides:** Four electron domains, one of them a lone pair.

A	Adjacent category	That is the electron-domain geometry, not the shape of the atoms.
B	Correct	Tetrahedral electron geometry minus one position gives a pyramid (about 107°).
C	Confusion set	Bent needs two lone pairs and two bonds, as in water.
D	Dropped element	Ignores the lone pair.

5. Answer B Concept 3 · transfer

A central atom has four bonding pairs and two lone pairs (AX₄E₂), as in XeF₄. What is the molecular shape?

Target word: shape from domain count **Decides:** Six domains; lone pairs opposite each other.

A	Adjacent category	Electron geometry, not molecular shape.
B	Correct	Octahedral electron geometry with two trans lone pairs.
C	Confusion set	Seesaw is AX ₄ E, with one lone pair.
D	Dropped element	Counts only the four bonds.

6. Answer C Concept 4

A specialty-gas firm ~~founded by two engineers~~ runs the reaction $\text{BF}_3 + \text{NH}_3 \rightarrow \text{F}_3\text{B}-\text{NH}_3$. Which species acts as the Lewis acid?

Target word: Lewis acid **Decides:** Boron has an empty orbital and accepts nitrogen's lone pair.

A	Confusion set	One donates and one accepts; roles differ.
B	Adjacent category	Applies the Brønsted definition; Lewis acids need no proton.
C	Correct	Electron-pair acceptor.
D	Inverted operation	NH ₃ donates the pair; it is the Lewis base.

7. Answer D Concept 5

A pigment maker ~~whose products are sold in 40 countries~~ uses K₄[Fe(CN)₆]. Cyanide is CN[−]. What are iron's oxidation state and coordination number?

Target word: oxidation state and CN **Decides:** Complex charge −4; six CN[−] ligands give −6.

A	Confusion set	That is iron in ferricyanide, [Fe(CN) ₆] ^{3−} .
B	Wrong stem number	Uses the four K ⁺ counter-ions as the coordination number.
C	Adjacent category	Counts one positive charge per ligand instead of balancing charges.
D	Correct	$\text{Fe} + 6(-1) = -4$, so Fe = +2; six donor atoms.

8. Answer D Concept 5 · transfer

In [Co(NH₃)₅Cl]Cl₂, cobalt is +3. What is the charge on the complex ion in brackets?

Target word: solve for complex charge **Decides:** +3 + 5(0) + (−1).

A	Dropped element	Drops the coordinated chloride.
B	Confusion set	That is the whole neutral compound.
C	Adjacent category	Treats each neutral NH ₃ as +1: 3 + 5 − 1.
D	Correct	The inner chloride is −1; two outer Cl [−] balance +2.

9. Answer A Concept 6

A mine's water-treatment plant, ~~which handles 2 million gallons a day,~~ removes iron with $\text{MnO}_4^- + 5\text{Fe}^{2+} + 8\text{H}^+ \rightarrow \text{Mn}^{2+} + 5\text{Fe}^{3+} + 4\text{H}_2\text{O}$. Which species is the oxidizing agent?

Target word: oxidizing agent **Decides:** Mn falls from +7 to +2 (reduced).

A	Correct	It is reduced, so it oxidizes Fe^{2+} .
B	Inverted operation	Fe^{2+} is oxidized; it is the reducing agent.
C	Adjacent category	A product, the reduced form of the agent.
D	Confusion set	Hydrogen stays +1; it balances oxygen, not electrons.

10. Answer A Concept 6 · transfer

In the same reaction, manganese goes from +7 to +2. How many electrons does each manganese gain?

Target word: electrons transferred **Decides:** Change in oxidation state = 5.

A	Correct	+7 to +2 is a gain of 5 electrons.
B	Inverted operation	Adds the states instead of subtracting.
C	Wrong stem number	The final oxidation state, not the change.
D	Dropped element	The starting state; drops the endpoint.

11. Answer C Concept 7

A research group ~~on a three-year grant~~ studies $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, a high-spin octahedral d5 complex. How many unpaired electrons does it have, and what is its magnetism?

Target word: unpaired electrons **Decides:** High spin: one electron in each of five d orbitals.

A	Dropped element	Counts only the t2g set and drops the eg electrons.
B	Confusion set	An odd electron count cannot be fully paired.
C	Correct	t2g ³ eg ² with no pairing.
D	Adjacent category	That is the low-spin d5 case.

12. Answer D Concept 8

A teacher ~~whose school just remodeled its chemistry lab~~ drops small pieces of lithium, sodium, and potassium into water behind a safety shield. Which reacts most vigorously, and why?

Target word: most reactive alkali metal **Decides:** Lowest ionization energy down the group.

A	Inverted operation	Reverses the trend.
B	Absolute language	Groups share patterns, not identical reactivity.
C	Confusion set	Middle position is not the extreme.
D	Correct	Alkali reactivity rises down the group.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	11	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Periodic trends	Atomic radius decreases across a period and increases down a group; first ionization energy runs the opposite way.	Thinking more electrons across a period means a bigger atom.	Elements in the same row or column ranked by size or ionization energy.
2	Ionic compound properties	Ionic solids have high melting points and conduct electricity only when melted or dissolved, when ions can move.	Expecting solid salts to conduct like metals.	A metal-nonmetal compound with a property question.
3	VSEPR shapes with lone pairs	Electron-domain geometry counts bonds and lone pairs; molecular shape names only atom positions.	Naming the electron geometry as the molecular shape, or ignoring lone pairs.	A central atom with lone pairs (NH ₃ , H ₂ O, XeF ₄).
4	Lewis acids and bases	A Lewis acid accepts an electron pair; a Lewis base donates one. No proton transfer is needed.	Requiring an H ⁺ transfer to call something an acid.	An adduct forming between an electron-poor and a lone-pair species.
5	Coordination compounds	Metal oxidation state = complex charge – sum of ligand charges; coordination number = number of donor atoms bound to the metal.	Ignoring ligand charges or counter-ions.	A bracketed complex with counter-ions.
6	Redox agents	The oxidizing agent is reduced (gains electrons, oxidation state falls); the reducing agent is oxidized.	Calling the species that is oxidized the oxidizing agent.	Oxidation states that change across a reaction.
7	Crystal field and magnetism	In octahedral complexes, weak-field ligands give high-spin (max unpaired) and strong-field give low-spin; unpaired electrons make a complex paramagnetic.	Filling only the lower t _{2g} set regardless of spin state.	“High-spin” or “low-spin” with a d-electron count.
8	Group reactivity trends	Alkali-metal reactivity increases down the group as the outer electron is held more loosely; halogen oxidizing strength decreases down the group.	Assuming elements in a group react identically, or that smaller atoms react faster.	Members of one group compared in a reaction.

Course scope (public description): Inorganic Chemistry introduces the concepts of inorganic chemistry—the branch of chemistry that studies the properties and behavior of any compound, avoiding a specific focus on carbon. It will focus on the three most important areas of inorganic chemistry: the structure, properties, and reactions of various groups of inorganic compounds.